

5.4 Gravitational fields

This section provides knowledge and understanding of Newton's law of gravitation, planetary motion and gravitational potential and energy.

Learners have the opportunity to appreciate how theories are used to predict the geostationary orbit of artificial satellites and the escape velocity of atoms in the atmosphere of planets (HSW1 and 2).

5.4.1 Point and spherical masses

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) gravitational fields are due to objects having mass	
(b) modelling the mass of a spherical object as a point mass at its centre	
(c) gravitational field lines to map gravitational fields	
(d) gravitational field strength; $g = \frac{F}{m}$.	